

GRASP: Graph Representation Learning with Assay Supervision for Molecular Properties

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Abstract

Molecular structures are abundant, while experimental bioactivity is sparse and distributed across assays. We introduce GRASP, a 93.5M-parameter graph Transformer that learns from these sources in sequence. GRASP first learns molecular structure through replaced-token detection on 1.54 billion ZINC20 molecule presentations, then adapts to sparse bioactivity across 642 ChEMBL assays. Across 23 cluster-held-out OpenADMET endpoints, GRASP achieves a mean MAE of 0.374, compared with 0.383 for CheMeleon, 0.408 for ECFP4-LightGBM, and 0.440 for Chemprop. ChEMBL adaptation reduces mean MAE from 0.407 to 0.374 and improves 21 of 23 endpoints, showing the value of intermediate bioactivity supervision. On 22 TDC ADMET tasks, GRASP gives a better point estimate than CheMeleon on 13 tasks. These results support staged molecular representation learning, where abundant molecular structure and sparse experimental measurements provide complementary supervision at different stages of training.

Introduction

Learning useful molecular representations is difficult because we have far more data about molecular structures than about experimentally measured properties [1, 2]. Databases such as ZINC contain enormous numbers of molecular structures [3], but experimental property data is available for only a small fraction of these molecules. Obtaining it is expensive because each measurement requires laboratory experiments, so the data remains sparse. It is also spread across many different assays, each covering a different set of molecules, with little overlap between them [4]. Most molecular pretraining pipelines use the first resource for self-supervision and then move directly to endpoint-specific fine-tuning. This leaves a valuable source of supervision underutilised: large bioactivity databases such as ChEMBL, whose incomplete assay matrices still connect molecular structure to experimentally observed chemical behaviour [5, 6].

We introduce Graph Representation Learning with Assay Supervision for Molecular Properties - GRASP, a 93.5M-parameter graph Transformer that learns from these sources in sequence. The first stage learns molecular structure through replaced-token detection (RTD): a compact generator proposes contextual atom-token replacements, and a discriminator predicts which atoms were replaced [7]. Each atom is represented by a radius-0 Morgan token, while shortest-path relative attention supplies molecular topology [8–10]. Structural training uses 1.54B molecular presentations from ZINC20. The second stage adapts the encoder using approximately 512,000 molecules and 642

sparsely observed ChEMBL assays. The resulting model can then be adapted to predict a specific molecular property by full fine-tuning, LoRA, or frozen layer mixing [11].

On 23 cluster-held-out OpenADMET endpoints [12–14], full fine-tuning reaches a mean MAE of 0.374, compared with 0.383 for CheMeleon, 0.408 for tuned ECFP4-LightGBM, and 0.440 for Chemprop D-MPNN. GRASP has lower endpoint MAE than CheMeleon on 12 of 23 endpoints, ECFP4-LightGBM on 16 of 23, and Chemprop on 22 of 23. With frozen encoders, GRASP also improves on Mol-JEPA on 18 of 23 endpoints under the same splits and regression protocol.

In the matched stage comparison, ChEMBL adaptation lowers mean MAE from 0.407 to 0.374 and improves 21 endpoints, in GRASP-Small, RTD improves on the implemented masked-token objective on 16 endpoints. We further evaluate the model on the 22-task TDC ADMET benchmark, which covers both classification and regression tasks [15]. Overall, the results show that learning molecular structure at scale and then adapting on sparse experimental data improves performance on downstream property prediction.

Related Work

Controlled studies show that molecular pretraining gains depend on features, architecture, data splits, and downstream optimization [16], while scaling studies report benefits from larger and more diverse supervision [17, 18]. Molecular representation learning has consequently explored diverse signals, including self-supervised graph objectives (GROVER) [19], contrastive learning (MolCLR) [20], molecular strings (ChemBERTa, MoLFormer) [21, 22], geometry (Uni-Mol/Uni-Mol2) [23, 24], descriptors (CheMeleon) [25], and bioactivity supervision (MolE) [26]. GRASP instead combines shortest-path relative attention [9, 10] with ELECTRA-style replaced-token detection and gradient-disentangled embedding sharing [7, 27]; replacement-based graph objectives have also been studied in DiscoGNN [28]. Rather than attributing performance to individual architectural components, we test two controlled questions: whether RTD improves over masked-token pretraining, and how sparse multi-assay bioactivity adaptation changes the resulting representation.

Overview of GRASP

Molecular representation and encoder. We tokenize each molecule with RDKit radius-0 Morgan hashes, yielding one atom token from a 211-token vocabulary including PAD, MASK, UNK, and CLS [8, 29]. The input also contains a relative-distance index $r(i, j)$ derived from shortest-path bond distance. For connected atom pairs, $r(i, j) = \delta(i, j) + 1$, where $\delta(i, j)$ is their shortest-path bond distance. We set $r(i, j) = 0$ for pairs involving CLS or padding and for atom pairs in disconnected components. The discriminator is a 12-layer graph Transformer with hidden size 768, 12 heads, feed-forward size 3,072, and 93,484,800 encoder parameters [30]. It uses content-to-position relative attention,

$$a_{ij} = \frac{\mathbf{q}_i^\top \mathbf{k}_j + \mathbf{q}_i^\top \mathbf{k}_{r(i,j)}^{\text{rel}}}{\sqrt{2d_h}}, \quad d_h = 64, \quad (1)$$

together with learned absolute positional embeddings based on the RDKit atom ordering, while the attention scores themselves remain content-based.

Graph replaced-token detection. Figure 1 summarizes the objective.

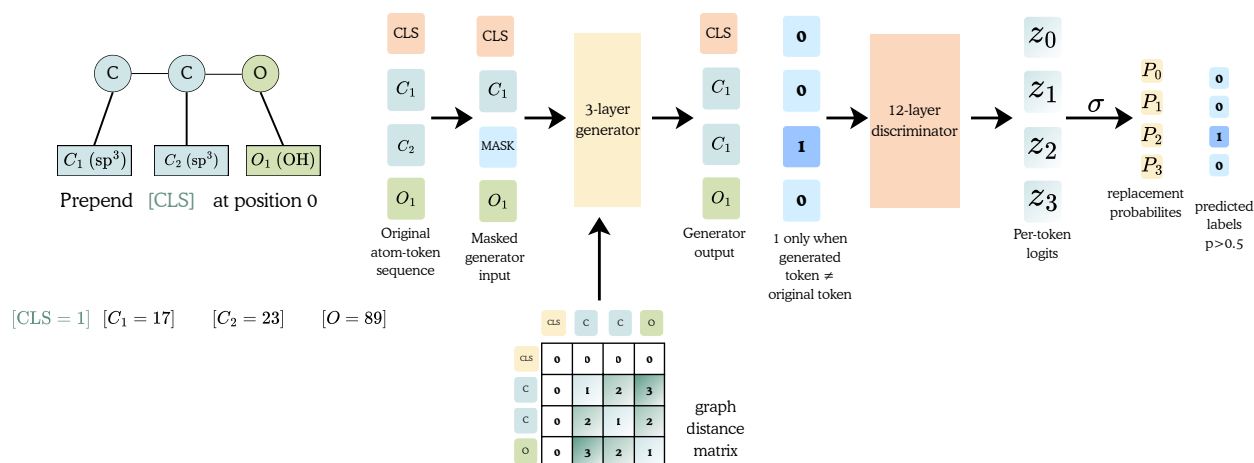


Figure 1. Graph replaced-token detection. Each heavy atom is tokenized as a radius-0 Morgan hash and a CLS token is prepended at position 0. Atom tokens are independently selected with probability 0.25 and replaced by MASK, and a three-layer generator predicts the original token at each selected position. The generator’s argmax prediction is written back into the graph, so a position is labelled as replaced only when the generated token differs from the original. The 12-layer discriminator receives the resulting token sequence together with the shortest-path graph-distance matrix and predicts, at every valid position, whether that token was replaced.

Each atom token is independently selected with probability 0.25 and replaced by MASK. A three-layer, 256-hidden generator predicts the original token, and its argmax prediction is inserted into the graph. For selected positions M , generator cross-entropy is

$$\mathcal{L}_G = -\frac{1}{|M|} \sum_{i \in M} \log p_G(x_i | \mathbf{x}_{\setminus M}, G). \quad (2)$$

The discriminator predicts at every valid position T , including CLS, whether the realized input differs from the original: $r_i = \mathbb{I}[\tilde{x}_i \neq x_i]$, with $r_{\text{CLS}} = 0$. Its loss is

$$\mathcal{L}_D = -\frac{1}{|T|} \sum_{i \in T} [r_i \log D_i + (1 - r_i) \log(1 - D_i)], \quad \mathcal{L} = \mathcal{L}_G + 50\mathcal{L}_D. \quad (3)$$

Gradient-disentangled embedding sharing implements $W_D = \text{stopgrad}(W_G) + B_D$: the generator supplies shared token-embedding values, while discriminator gradients update only the learned bias B_D .

Two-stage training. Step 1 applies RTD to 1,538,240,768 molecular presentations from 1,538,240,669 ZINC20 records over 6,133,753 optimizer updates [3]. Step 2 fine-tunes the encoder for 80,000 updates on 511,898 ChEMBL 36 molecules and 642 binary binding or functional assays [5, 6]. Labels come from activity comments or pChEMBL thresholds (≥ 5.5 active, ≤ 4.5 inactive); contradictory pairs are removed and assays require at least 500 molecules. A linear 642-output head reads the final-layer CLS state, and binary cross-entropy is computed only for observed labels. The fixed 50,000-step encoder snapshot is used downstream.

Downstream adaptation. For OpenADMET, the three adaptation modes use a softmax-weighted mixture of layers 4, 8, 10, and 12, additive attention pooling over atom tokens (excluding CLS),

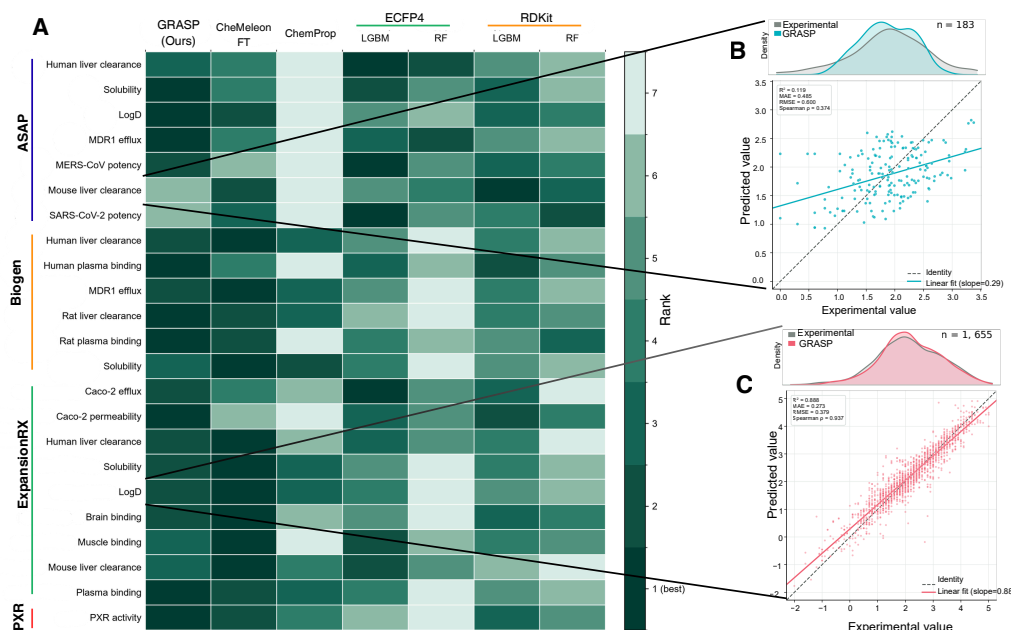


Figure 2. OpenADMET-23 performance. (A) Rank of each method on each of the 23 cluster-held-out endpoints, from 1 (best, dark) to 7, with endpoints grouped by source. Ranks are computed from the endpoint mean MAE over the three splits reported in Table A2. (B, C) Predicted against experimental values for GRASP full fine-tuning on two endpoints, pooled over the three splits, illustrating the range of difficulty in the panel: (B) ASAP mouse liver clearance ($n = 183$), among the weakest endpoints for every method, and (C) ExpansionRx LogD ($n = 1,655$), among the strongest. Marginal densities compare the experimental and predicted distributions. The dashed line is identity and the solid line is the least-squares fit; pooled MAE differs slightly from the split-averaged MAE in Table A2 because splits have unequal test sizes.

and a 512-hidden MLP head. Full fine-tuning updates the entire encoder; frozen mixing keeps the encoder fixed; LoRA [31] keeps the base weights fixed and adds rank-16 query/value updates ($\alpha = 32$, dropout 0.1) in every attention layer. The secondary TDC evaluation uses a separate task-specific protocol based on the final-layer CLS representation, described in Appendix ??.

Results

GRASP improves molecular property prediction across diverse ADMET endpoints

Task: We evaluate molecular property prediction on OpenADMET, a multi-endpoint benchmark spanning clearance, solubility, permeability, plasma and tissue binding, efflux, PXR activity, and antiviral Mpro potency (pIC50 or pEC50). Given a molecular graph, the model predicts the continuous value of each endpoint independently. Performance is measured using mean absolute error (MAE), with lower values indicating better predictive performance.

Dataset and Evaluation: OpenADMET comprises 23 endpoints, with 43,881 labelled measurements across 14,261 unique SMILES. Each endpoint is evaluated on three reconstructed cluster-held-out splits, and we report the test MAE averaged across the three splits. Dataset sizes, target transformations, and endpoint-specific details are provided in Appendix Table A4.

Baselines: We compare full fine-tuning (FT) with ECFP4-LightGBM, RDKit-LightGBM,

Table 1. OpenADMET-23 summary. % impr. is the relative reduction in mean MAE achieved by GRASP full fine-tuning (FT) over the comparator, $(1 - \text{MAE}_{\text{FT}}/\text{MAE}_{\text{comp}}) \times 100$, and Wins/23 counts endpoints where FT has lower MAE. In the frozen-encoder block, the comparison is instead GRASP frozen mixing versus Mol-JEPA. All comparisons use identical reconstructed cluster-held-out splits.

Comparator	Mean MAE	% impr.	Wins/23
<i>Rerun baselines (vs. FT, 0.374 ± 0.003)</i>			
ECFP4-LightGBM	0.408 ± 0.014	8.2	16
RDKit-LightGBM	0.412 ± 0.011	9.2	21
RDKit random forest	0.431 ± 0.001	13.1	21
Chemprop D-MPNN	0.440 ± 0.016	14.9	22
ECFP4 random forest	0.453 ± 0.001	17.3	20
CheMeleon FT	0.383 ± 0.008	2.4	12
<i>Adaptation of GRASP (vs. FT)</i>			
LoRA	0.380	1.6	14
Frozen layer mixing	0.387	3.4	18
<i>Frozen encoder (vs. GRASP frozen, 0.3869)</i>			
Mol-JEPA (CLS + MLP)	0.4123	6.2	18

Table 2. OpenADMET adaptation comparison. Mean MAE averages the 23 endpoint means across three cluster-held-out splits; $\pm$ denotes population standard deviation across split-wise macro-averages where available. Pairwise p -values compare each method with full fine-tuning and are Holm-adjusted across all three adaptation contrasts. The LoRA–frozen contrast is included in the correction but is not shown. The overall Friedman test gives $\chi^2(2) = 7.91$, $p = 0.019$.

Method	Mean MAE $\downarrow$	p_{Holm} vs. FT
Full fine-tuning	0.374 ± 0.003	—
LoRA	0.380	0.268
Frozen layer mixing	0.387	0.037

ECFP4 and RDKit random forests, Chemprop, and CheMeleon, all evaluated on identical partitions. For the frozen-encoder comparison, we use each model’s architecture-native molecular representation: Mol-JEPA’s released CLS embedding and GRASP’s learned mixture of layers 4, 8, 10, and 12 followed by atom-attention pooling. Both representations are evaluated using the same regression MLP and the same data partitions and downstream training protocol. Thus, the downstream evaluation is matched, while the representation readout remains architecture-specific.

Performance: GRASP full fine-tuning reaches a mean MAE of 0.374 ± 0.003 across the 23 endpoints, the lowest of the methods compared (Table 1). It improves on Chemprop D-MPNN by 14.9% (22 of 23 endpoints) and on tuned ECFP4-LightGBM by 8.2% (16 of 23). The margin over CheMeleon, the strongest baseline, is smaller: 2.4% in the mean, with lower MAE on 12 of 23 endpoints. Per-endpoint results for all methods are given in Table A2 and paired contrasts in Table A3; Figure 2 shows the corresponding per-endpoint ranks and two representative fits. GRASP FT also had the lowest mean MAE among adaptation strategies (FT: 0.374; LoRA: 0.380; frozen mixing: 0.387), with evidence of a difference only against frozen mixing after correction ($p = 0.037$). Under this common downstream evaluation protocol, GRASP also improved over Mol-JEPA (0.387 vs. 0.412; 18/23 endpoints; $\Delta\text{MAE} = -0.025$, 95% CI $[-0.042, -0.008]$).

RTD provides gains over MLM for pretraining

Task: We compare replaced-token detection (RTD) with masked language modeling (MLM) for GRASP pretraining. Both models use the same encoder architecture, pretraining data, ChEMBL adaptation stage, and downstream evaluation. The models differ only in the pretraining objective. RTD detects replaced radius-0 atom tokens. MLM predicts masked radius-2 atom-environment tokens.

Dataset and Evaluation: Training a second model at the full 1.538B-presentation scale requires substantial compute. We therefore compare RTD and MLM with the smaller GRASP-Small configuration. Both models use 415M ZINC molecule presentations and the same 12-layer encoder with a hidden size of 768. The encoder has approximately 89M parameters. Both models use a 15% selected-token rate. Both models then use the same ChEMBL adaptation stage with 641 assays and 80,000 updates.

We freeze both encoders for the downstream evaluation. We evaluate them on 23 OpenADMET endpoints with the same three data splits. Both models use the same layer-mixing readout and regression head. Table 3 reports the test MAE averaged across the three splits.

Performance: RTD achieves lower mean MAE than GRASP-MLM (0.418 vs. 0.431), outperforming it on 16/23 endpoints as presented in Table 3. The paired mean improvement is 0.0122 MAE (Holm-adjusted exact Wilcoxon $p = 0.0969$). GRASP-RTD wins all five LogD and solubility endpoints, with the largest gains on Biogen solubility (+16.5%), ASAP LogD (+11.6%), and SARS-CoV-2 potency (+10.3%). MLM’s seven wins are all clearance or binding endpoints, largest on ASAP human liver clearance (−10.8%) and Biogen rat plasma binding (−9.9%).

ChEMBL bioactivity adaptation improves molecular property prediction

Task: We test the effect of the ChEMBL bioactivity adaptation stage. GRASP-S1 is the Step-1 checkpoint trained with structural RTD only. GRASP also completes Step-2 adaptation on 642 sparsely observed ChEMBL assays. Step 2 includes both additional training on ChEMBL molecules and supervision from measured assay values. Therefore, this comparison measures the effect of the complete adaptation stage rather than assay supervision alone.

Dataset and Evaluation: We fully fine-tune both models on the same 23 OpenADMET endpoints. We evaluate each endpoint on the same three cluster-held-out splits. We report the test MAE averaged across the three splits.

Performance: ChEMBL adaptation lowers the mean MAE from 0.4069 to 0.3742, an 8.0% reduction. GRASP gives a lower MAE on 21 of 23 endpoints Table 4). The largest relative gains occur for ASAP solubility (24.2%), ExpansionRx plasma binding (19.5%), ASAP LogD (16.4%), and ExpansionRx brain binding (14.3%). GRASP-S1 gives a lower MAE on two endpoints: ASAP mouse liver clearance (6.7%) and ExpansionRx muscle binding (1.0%).

Secondary evaluations: TDC

All necessary information about TDC is present in Appendix

Table 3. Frozen-transfer comparison of RTD and MLM recipes on OpenADMET-23. RTD-15%-S2 and MLM-15%-S2 both use GRASP-Small with a 15% selected-token rate. Both use radius-0 input tokens; MLM predicts radius-2 environments. After ChEMBL adaptation, both encoders are frozen and evaluated with the same learned layer-mixing readout and regression head. Values are mean MAE $\pm$ population standard deviation across three split-specific fits. Bold marks the lower unrounded mean. $\Delta = 100(\text{MAE}_{\text{MLM}} - \text{MAE}_{\text{RTD}})/\text{MAE}_{\text{MLM}}$; positive values favor RTD.

Endpoint	GRASP-RTD	GRASP-MLM	Δ (%)
ASAP			
Human liver clearance	0.38 $\pm$.04	0.34 $\pm$.10	-10.81
Solubility	0.39 $\pm$.04	0.42 $\pm$.13	+6.36
LogD	0.59 $\pm$.06	0.67 $\pm$.15	+11.59
MDR1 efflux	0.37 $\pm$.02	0.39 $\pm$.02	+5.30
MERS-CoV potency	0.60 $\pm$.08	0.60 $\pm$.12	+0.66
Mouse liver clearance	0.44 $\pm$.04	0.43 $\pm$.06	-2.68
SARS-CoV-2 potency	0.64 $\pm$.09	0.71 $\pm$.12	+10.32
Biogen			
Human liver clearance	0.37 $\pm$.03	0.38 $\pm$.00	+4.24
Human plasma binding	0.46 $\pm$.16	0.45 $\pm$.18	-1.79
MDR1 efflux	0.42 $\pm$.01	0.43 $\pm$.01	+2.91
Rat liver clearance	0.42 $\pm$.02	0.43 $\pm$.07	+2.19
Rat plasma binding	0.54 $\pm$.07	0.49 $\pm$.05	-9.84
Solubility	0.35 $\pm$.03	0.42 $\pm$.05	+16.50
ExpansionRx			
Caco-2 efflux	0.24 $\pm$.01	0.26 $\pm$.01	+6.06
Caco-2 permeability	0.35 $\pm$.04	0.35 $\pm$.05	+1.45
Human liver clearance	0.39 $\pm$.02	0.40 $\pm$.01	+1.16
Solubility	0.51 $\pm$.05	0.51 $\pm$.04	+1.34
LogD	0.42 $\pm$.04	0.42 $\pm$.02	+0.96
Brain binding	0.28 $\pm$.01	0.30 $\pm$.03	+6.93
Muscle binding	0.25 $\pm$.04	0.25 $\pm$.04	-0.31
Mouse liver clearance	0.38 $\pm$.02	0.38 $\pm$.03	-0.10
Plasma binding	0.32 $\pm$.03	0.30 $\pm$.05	-5.63
PXR			
PXR activity	0.55 $\pm$.04	0.58 $\pm$.03	+6.48
Mean	0.42 $\pm$.01	0.43 $\pm$.01	+2.83

GRASP remains competitive across TDC ADMET tasks

Task: We evaluate GRASP on 22 TDC ADMET tasks (13 classification, 9 regression) covering absorption, distribution, metabolism, excretion, and toxicity.

Dataset and Evaluation: We use the fixed test sets of the TDC ADMET benchmark group without applying an additional heavy-atom filter or molecular truncation. We select hyperparameters with five independently seeded scaffold splits of the benchmark training and validation data. Test labels remain unused during hyperparameter selection. After hyperparameter selection, we fully fine-tune GRASP in three independent training runs.

Baseline: We compare with the released CheMeleon D-MPNN results.

Performance: GRASP has the better point estimate on 13 and CheMeleon on 9 (Table 5). GRASP’s largest margins are half-life (0.455 vs. 0.325 Spearman), VDss (0.665 vs. 0.546), and bioavailability (0.687 vs. 0.579 AUROC). CheMeleon’s clearest lead is hERG (0.867 vs. 0.773 AUROC).

Table 4. Effect of ChEMBL bioactivity adaptation on OpenADMET-23. GRASP-S1 uses structural pretraining; GRASP additionally includes ChEMBL adaptation. Both checkpoints are fully fine-tuned on the same three cluster-held-out splits. Entries report mean test MAE $\pm$ population standard deviation across split-specific fits. The mean-row deviation is calculated from the three split-wise macro-average MAEs. Bold marks the lower unrounded mean. $\Delta = 100(\text{MAE}_{\text{S1}} - \text{MAE}_{\text{GRASP}})/\text{MAE}_{\text{S1}}$; positive values favor GRASP.

Endpoint	GRASP-S1	GRASP	Δ (%)
ASAP			
Human liver clearance	0.38 $\pm$.07	0.36 $\pm$.07	+7.66
Solubility	0.42 $\pm$.08	0.32 $\pm$.05	+24.21
LogD	0.56 $\pm$.14	0.47 $\pm$.06	+16.35
MDR1 efflux	0.35 $\pm$.03	0.32 $\pm$.03	+6.35
MERS-CoV potency	0.58 $\pm$.11	0.56 $\pm$.11	+4.43
Mouse liver clearance	0.46 $\pm$.04	0.49 $\pm$.05	-6.71
SARS-CoV-2 potency	0.66 $\pm$.03	0.62 $\pm$.05	+6.50
Biogen			
Human liver clearance	0.37 $\pm$.03	0.33 $\pm$.01	+12.34
Human plasma binding	0.52 $\pm$.11	0.47 $\pm$.13	+9.25
MDR1 efflux	0.37 $\pm$.03	0.34 $\pm$.02	+9.56
Rat liver clearance	0.41 $\pm$.02	0.38 $\pm$.02	+8.54
Rat plasma binding	0.47 $\pm$.03	0.45 $\pm$.06	+4.65
Solubility	0.38 $\pm$.03	0.37 $\pm$.00	+2.67
ExpansionRx			
Caco-2 efflux	0.25 $\pm$.02	0.23 $\pm$.01	+7.68
Caco-2 permeability	0.32 $\pm$.02	0.29 $\pm$.02	+8.14
Human liver clearance	0.38 $\pm$.02	0.35 $\pm$.02	+6.93
Solubility	0.44 $\pm$.03	0.42 $\pm$.02	+5.13
LogD	0.30 $\pm$.02	0.27 $\pm$.02	+9.08
Brain binding	0.29 $\pm$.01	0.25 $\pm$.01	+14.32
Muscle binding	0.26 $\pm$.04	0.26 $\pm$.01	-0.97
Mouse liver clearance	0.37 $\pm$.01	0.34 $\pm$.03	+7.44
Plasma binding	0.30 $\pm$.02	0.24 $\pm$.03	+19.53
PXR			
PXR activity	0.52 $\pm$.03	0.49 $\pm$.03	+5.03
Mean	0.41 $\pm$.01	0.37 $\pm$.00	+8.03

RTD improves performance across TDC tasks

Task and Evaluation: We compare RTD with MLM on TDC in the GRASP-Small configuration of section (415M ZINC molecules, 15% selected-token rate, 80,000-update ChEMBL stage). RTD detects replaced radius-0 tokens, whereas MLM predicts masked radius-2 environments, so this compares implemented recipes rather than a single loss. Scaffold partitions, downstream heads, training settings, and three seeds are identical.

Performance: RTD has the better point estimate on 17 of 22 tasks and MLM on 5 (exacttwo-sided sign test, raw $p = 0.0169$, Holm-adjusted $p = 0.0338$; Table A12). RTD’s gains include microsomal clearance (0.589 vs. 0.467 Spearman), hERG (0.818 vs. 0.753 AUROC), and lipophilicity (0.459 vs. 0.505 MAE). MLM’s five wins are the three CYP substrate tasks, DILI, and P-glycoprotein. Complete task-level results are reported in Table A12.

ChEMBL bioactivity adaptation improves transfer across TDC tasks

Task: We test the effect of the ChEMBL bioactivity adaptation stage on downstream transfer. GRASP-S1 contains only the structural RTD stage. GRASP also includes the ChEMBL adaptation

Table 5. GRASP on the 22-task TDC ADMET benchmark. GRASP values are mean $\pm$ population standard deviation over three seeds after validation-only hyperparameter selection.

Task	Metric	GRASP	CheMeleon
Absorption			
Caco-2 Wang	MAE $\downarrow$	0.39 $\pm$.01	0.38 $\pm$.03
Bioavailability Ma	AUROC $\uparrow$	0.69 $\pm$.02	0.58 $\pm$.02
Lipophilicity AstraZeneca	MAE $\downarrow$	0.41 $\pm$.00	0.46 $\pm$.01
Solubility AqSolDB	MAE $\downarrow$	0.74 $\pm$.01	0.89 $\pm$.11
HIA Hou	AUROC $\uparrow$	0.95 $\pm$.02	0.96 $\pm$.01
P-glycoprotein Broccatelli	AUROC $\uparrow$	0.92 $\pm$.01	0.93 $\pm$.01
Distribution			
BBB Martins	AUROC $\uparrow$	0.88 $\pm$.02	0.89 $\pm$.02
PPBR AZ	MAE $\downarrow$	7.30 $\pm$.35	7.82 $\pm$.15
VDss Lombardo	Spearman $\uparrow$	0.67 $\pm$.02	0.55 $\pm$.01
Metabolism			
CYP2C9 inhibition	AUPRC $\uparrow$	0.79 $\pm$.00	0.78 $\pm$.01
CYP2D6 inhibition	AUPRC $\uparrow$	0.72 $\pm$.02	0.68 $\pm$.00
CYP3A4 inhibition	AUPRC $\uparrow$	0.89 $\pm$.01	0.87 $\pm$.01
CYP2C9 substrate	AUPRC $\uparrow$	0.41 $\pm$.03	0.43 $\pm$.01
CYP2D6 substrate	AUPRC $\uparrow$	0.64 $\pm$.03	0.67 $\pm$.04
CYP3A4 substrate	AUROC $\uparrow$	0.67 $\pm$.04	0.61 $\pm$.01
Excretion			
Half-life Obach	Spearman $\uparrow$	0.46 $\pm$.06	0.33 $\pm$.03
Clearance hepatocyte AZ	Spearman $\uparrow$	0.47 $\pm$.02	0.38 $\pm$.05
Clearance microsome AZ	Spearman $\uparrow$	0.60 $\pm$.02	0.60 $\pm$.02
Toxicity			
LD50 Zhu	MAE $\downarrow$	0.49 $\pm$.00	0.53 $\pm$.01
hERG	AUROC $\uparrow$	0.77 $\pm$.03	0.87 $\pm$.01
Ames	AUROC $\uparrow$	0.83 $\pm$.01	0.85 $\pm$.01
DILI	AUROC $\uparrow$	0.93 $\pm$.02	0.90 $\pm$.03
Favorable tasks		13/22	9/22

stage with sparse assay supervision. This comparison measures the effect of the complete adaptation stage rather than assay supervision alone.

Dataset and Evaluation: We evaluate both checkpoints on the same 22 TDC ADMET tasks. Both models use the same fixed test sets and downstream heads. We select hyperparameters with the same five scaffold splits of the training and validation data. Both models use the same three random seeds for downstream training.

Performance: Using the GRASP results reported in Table 5, GRASP has the favorable point estimate over GRASP-S1 on 19 of the 22 tasks. Representative gains include CYP3A4 substrate prediction (AUROC: 0.565 ± 0.046 to **0.674 ± 0.035**), CYP2D6 inhibition (AUPRC: 0.624 ± 0.009 to **0.716 ± 0.017**), hepatocyte clearance (Spearman: 0.408 ± 0.042 to **0.471 ± 0.018**), and microsomal clearance (Spearman: 0.556 ± 0.048 to **0.602 ± 0.017**). GRASP also reduces MAE on Caco-2 permeability (0.470 ± 0.040 to **0.389 ± 0.009**), solubility (0.794 ± 0.003 to **0.736 ± 0.009**), lipophilicity (0.449 ± 0.011 to **0.406 ± 0.003**), and LD50 (0.576 ± 0.002 to **0.486 ± 0.002**). GRASP-S1 retains the favorable point estimate on half-life, hERG, and HIA. Overall, the adapted GRASP checkpoint is favorable on most TDC endpoints, with gains spanning classification and regression tasks.

Discussion, Conclusion and Limitations

GRASP demonstrates that a molecular encoder can benefit from learning structure at scale and then incorporating sparse experimental bioactivity. On OpenADMET-23, full fine-tuning reaches a mean MAE of 0.3742, compared with 0.4075 for ECFP4-LightGBM and 0.4395 for Chemprop. The endpoint counts are important: GRASP improves on ECFP4-LightGBM on 16 of 23 tasks (Holm-adjusted sign test $p = 0.093$; the paired Wilcoxon test over the same endpoints gives $p = 0.006$) and on Chemprop on 22 of 23. The result is therefore not only a lower macro-average; the model is useful across most of the evaluated clearance, solubility, permeability, binding, efflux, and activity endpoints. GRASP is the single best method on 9 of the 23 endpoints (Table A2), tuned fingerprint and descriptor models remain competitive per endpoint at these training-set sizes, and the advantage we report is a consistent aggregate one. On the separate TDC benchmark, GRASP has the favorable point estimate on 13 of 22 tasks with a comparable CheMeleon value, extending the evaluation to both classification and regression under scaffold holdout.

The frozen-encoder experiment shows that this performance is not limited to updating the full backbone. Using the same endpoint partitions and regression MLP, frozen GRASP obtains a mean MAE of 0.3869, compared with 0.4123 for Mol-JEPA, and improves on 18 of 23 endpoints. Downstream adaptation provides an additional gain: full fine-tuning reaches 0.3742, LoRA reaches 0.3800, and frozen layer mixing reaches 0.3869. LoRA updates only 1.04% of the parameters, making it a strong option when many endpoint-specific models must share one pretrained encoder. The difference between full fine-tuning and LoRA is not resolved after multiplicity correction, whereas full fine-tuning is separated from frozen mixing.

The controlled training comparisons support both stages of the recipe. In GRASP-Small, RTD improves on the implemented MLM objective on 16 of 23 OpenADMET endpoints and 17 of 22 TDC tasks. Adding sparse ChEMBL bioactivity adaptation lowers OpenADMET mean MAE from 0.4069 to 0.3742 and improves 21 of 23 endpoints relative to the Step-1 checkpoint. Individual endpoints do not always favor GRASP, but the main pattern is consistent across the primary comparison, the frozen-encoder analysis, the stage comparison, and the secondary benchmark. GRASP therefore provides a practical foundation for molecular property prediction: one graph encoder, trained from abundant structures and sparse assays, that supports both accuracy-oriented full fine-tuning and parameter-efficient adaptation.

This ability to adapt from limited experimental supervision could also support prospective molecular discovery, including low-data assay modelling and active-learning [32] or lab-in-the-loop workflows, where the encoder is continually updated as new measurements become available to prioritize subsequent compounds for evaluation.

Limitations. Several conclusions are intentionally narrower than the full training recipe. First, the GRASP-S1-GRASP comparison evaluates the complete ChEMBL adaptation stage and does not isolate assay supervision from additional optimization or exposure to the ChEMBL molecular distribution. Second, the structural-overlap analysis tests molecular exposure but does not establish the absence of semantically related assay supervision between ChEMBL and downstream endpoints. Third, the RTD-MLM comparison is performed in GRASP-Small and compares implemented pretraining recipes rather than isolating pretraining scale. Fourth, the principal downstream experiments use the 50,000-step Step-2 snapshot without a systematic downstream checkpoint-sensitivity study. Finally, the TDC comparison with released CheMeleon scores is descriptive because the two methods were not trained and selected within a single paired experimental pipeline.

AI Use Disclosure

Generative AI tools were used to assist with manuscript writing and editing. All AI-assisted content was reviewed and verified by the authors, who take full responsibility for the final work.

Code and Model Availability

The implementation and model-release information are available at <https://github.com/caithmac/GRASP>.

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Additional Experimental Details

Architecture and Training Details

This appendix specifies the input representation, architecture, and training settings needed to reproduce the model.

RDKit uses radius 0, `useFeatures=False`, redundant environments enabled, and chirality disabled. The 32-bit hashes form 207 observed atom types plus PAD, MASK, UNK, and CLS (211 entries); unseen hashes map to UNK. These hashes are not reversible feature tuples. For connected atom pairs, graph distance d is stored as the relative-distance index $d + 1$; index 0 is reserved for pairs involving CLS or padding and for atom pairs in disconnected components. The implementation’s `edge_attr` tensor stores distance indices rather than chemical bond features.

ChEMBL binding and functional labels are derived from activity comments or pChEMBL: values ≥ 5.5 are active, values ≤ 4.5 inactive, and intermediate values are discarded. Contradictory molecule–assay pairs are removed, and each assay requires at least 500 molecules.

Table A1. Encoder and pretraining configuration.

Quantity	Value
Encoder parameters	93,484,800
Layers / hidden / heads	12 / 768 / 12
Feed-forward dimension	3,072
Atom vocabulary	211
Generator layers / hidden	3 / 256
Selected-token probability	0.25
Generator replacement	argmax, unconstrained
RTD loss weight λ	50
Relative attention	content-to-position
Absolute-position capacity	512
Relative-position embedding rows	1,024
Relative-distance bucketing	none
Step-1/2 heavy-atom cap	96
Position-biased input	enabled
Molecules > 96 heavy atoms in Step 1/2	filtered, not truncated
Downstream 96-atom filter	none
Shared attention keys	disabled
Relative embedding norm	none
Step-1 presentations	1,538,240,768
Step-2 molecules / assays	511,898 / 642
Evaluated Step-2 snapshot	50,000 steps

The Step-1 exposure count is reconstructed from training manifests. Because interrupted jobs restarted with a shuffled stream, 1.538B denotes molecular presentations rather than unique molecules. Training comprised 2,500,000 updates at global batch 256 (640M presentations), 500,000 resumed updates at global batch 192 (96M presentations), and 3,133,753 exposure-correcting continuation updates at global batch 256 (802,240,768 presentations), for 6,133,753 optimizer updates in total. The final continuation ran in mixed precision on four GPUs with per-device batch 32 and gradient accumulation 2. It used AdamW with peak learning rate 5×10^{-5} , weight decay 0.01, 10,000 warmup steps, cosine decay, and gradient-norm clipping at 1.0.

Step 2 used four-GPU distributed training with per-device batch 32, gradient accumulation 4, and global batch 512. Adam used learning rate 5×10^{-6} , zero weight decay, 10,000 steps of linear warmup followed by linear decay, dropout 0.1, and gradient-norm clipping at 1.0. Training continued to 80,000 steps; the 50,000-step encoder was fixed for principal downstream experiments before task aggregation. The retained records identify four GPUs for the final Step-1 continuation and Step 2 but do not preserve accelerator model, wall time, GPU-hours, energy, or complete hardware accounting for earlier Step-1 phases; we report these quantities as unavailable rather than

reconstructing them.

Step-2 Label Matrix and Snapshot Sensitivity

This appendix describes the sparsity and class balance of the bioactivity labels and compares the available training snapshots.

The final Step-2 matrix contains 511,898 molecules and 642 assays, or 328,638,516 possible molecule–assay cells. Of these, 4,536,721 are observed (1.3805% density): 1,097,029 positive (24.18%) and 3,439,692 negative (75.82%).

OpenADMET Data, Splits, and Evaluation

In source assay codes, HLM, MLM, and RLM mean human, mouse, and rat liver microsomes; MLM elsewhere denotes masked language modeling. The main results spell out the clearance assays to distinguish them.

Full endpoint results are reported in Table [A2](#).

Statistical analysis. All paired differences use GRASP minus comparator MAE. We bootstrap endpoint means with 100,000 resamples and use exact two-sided Wilcoxon and sign tests. The six baseline contrasts form one Holm family for each test; the three adaptation contrasts form another. Size and scaffold associations use endpoint averages and Holm correction over four associations per validation or test family.

The three outer assignments are stored as `split1–split3`. Within each outer-training partition, complete clusters are allocated to a 15% validation subset with seeds 7301, 7302, and 7303. Neural fit seeds are 9200–9202, 9300–9302, and 9400–9402 for frozen, LoRA, and full adaptation in the three splits. Training targets are standardized from inner-training rows only and predictions are inverse-transformed before MAE is computed. Table [A4](#) reports executed row counts, target transforms, and test sizes; raw MAEs therefore remain endpoint-specific quantities and are macro-averaged only after each endpoint is scored.

Classical baseline search. For each feature type, the random-forest grid uses 500 trees, maximum-feature choices `sqrt` or 0.33, and minimum leaf sizes 1, 2, or 4 (six configurations). The LightGBM grid uses 600 trees, 31 or 63 leaves, minimum child counts 10 or 20, and learning rates 0.03 or 0.06 (eight configurations), row subsampling is 0.9, feature subsampling 0.8, and L2 regularization 1.0. Validation MAE selects one configuration per endpoint and split. Chemprop 2.3.0 uses its fixed default D-MPNN architecture with a single model, no ensembling, batch size 64, a 60-epoch cap, and validation-loss early stopping with patience 12.

Molecule size handling. The 96-heavy-atom limit was applied during pretraining but was not imposed globally on OpenADMET. Downstream molecules were neither truncated nor filtered by size: all atom tokens were retained, and variable-length inputs were padded only to the longest molecule within each batch. The encoder has an absolute-position capacity of 512 tokens. Across the 43,881 OpenADMET endpoint rows, only one row exceeded the pretraining limit: a PXR molecule with 123 heavy atoms. This molecule occurred in the inner-training partition in all three PXR splits and never in a validation or test partition. Consequently, all reported OpenADMET test sets contain molecules with at most 96 heavy atoms.

Table A2. Per-endpoint MAE on OpenADMET-23 (lower is better). Each entry reports the mean $\pm$ population standard deviation across three cluster-held-out splits. These deviations measure split-to-split variation, not repeated-seed variation on a fixed split. CheMeleon was initialized from its released foundation checkpoint and fully fine-tuned. **Bold** marks the best mean and underlining marks the second-best mean per endpoint. Rankings are based on unrounded means. $\Delta = 100(\text{MAE}_{\text{competitor}} - \text{MAE}_{\text{GRASP}})/\text{MAE}_{\text{competitor}}$. The competitor is the runner-up when GRASP is best and the overall best method otherwise. Positive values favor GRASP; negative values favor the comparator. All methods were evaluated on identical partitions.

Endpoint	GRASP FT	CheMeleon FT	ECFP4-LGBM	RDKit-LGBM	Chemprop	Δ (%)
ASAP						
Human liver clearance	0.36 $\pm$.07	0.38 $\pm$.04	0.33 $\pm$.07	0.38 $\pm$.03	0.44 $\pm$.06	-7.2
Solubility	0.32 $\pm$.05	0.38 $\pm$.07	<u>0.33 $\pm$.07</u>	0.36 $\pm$.10	0.44 $\pm$.09	+4.3
LogD	0.47 $\pm$.06	<u>0.55 $\pm$.12</u>	0.70 $\pm$.19	0.66 $\pm$.17	0.83 $\pm$.24	+13.9
MDR1 efflux	0.32 $\pm$.03	0.35 $\pm$.06	<u>0.34 $\pm$.04</u>	0.38 $\pm$.02	0.44 $\pm$.04	+6.0
MERS-CoV potency	<u>0.56 $\pm$.11</u>	0.60 $\pm$.07	0.55 $\pm$.09	0.57 $\pm$.10	0.63 $\pm$.05	-0.5
Mouse liver clearance	0.49 $\pm$.05	0.46 $\pm$.03	0.48 $\pm$.08	0.46 $\pm$.07	0.51 $\pm$.04	-7.1
SARS-CoV-2 potency	0.62 $\pm$.05	<u>0.58 $\pm$.06</u>	0.55 $\pm$.02	0.58 $\pm$.05	0.64 $\pm$.06	-12.6
Biogen						
Human liver clearance	<u>0.33 $\pm$.01</u>	0.33 $\pm$.01	0.36 $\pm$.01	0.35 $\pm$.01	0.34 $\pm$.01	-0.6
Human plasma binding	0.47 $\pm$.13	0.49 $\pm$.12	0.48 $\pm$.17	<u>0.48 $\pm$.12</u>	0.61 $\pm$.24	+1.7
MDR1 efflux	<u>0.34 $\pm$.02</u>	0.33 $\pm$.04	0.37 $\pm$.02	0.37 $\pm$.01	0.36 $\pm$.02	-0.8
Rat liver clearance	0.38 $\pm$.02	<u>0.38 $\pm$.03</u>	0.47 $\pm$.03	0.42 $\pm$.03	0.41 $\pm$.02	+1.5
Rat plasma binding	0.45 $\pm$.06	<u>0.45 $\pm$.04</u>	0.47 $\pm$.03	0.50 $\pm$.01	0.51 $\pm$.03	+0.5
Solubility	0.37 $\pm$.00	0.34 $\pm$.01	0.38 $\pm$.04	0.39 $\pm$.05	<u>0.35 $\pm$.02</u>	-8.8
ExpansionRx						
Caco-2 efflux	<u>0.23 $\pm$.01</u>	0.24 $\pm$.01	0.23 $\pm$.00	0.23 $\pm$.02	0.25 $\pm$.01	-1.1
Caco-2 permeability	0.29 $\pm$.02	0.37 $\pm$.06	0.33 $\pm$.03	<u>0.32 $\pm$.03</u>	0.41 $\pm$.03	+9.4
Human liver clearance	0.35 $\pm$.02	0.35 $\pm$.00	0.35 $\pm$.00	0.37 $\pm$.01	0.37 $\pm$.00	-0.8
Solubility	<u>0.42 $\pm$.02</u>	0.41 $\pm$.04	0.53 $\pm$.10	0.53 $\pm$.08	0.43 $\pm$.04	-0.5
LogD	<u>0.27 $\pm$.02</u>	0.26 $\pm$.00	0.38 $\pm$.02	0.41 $\pm$.03	0.28 $\pm$.01	-4.1
Brain binding	<u>0.25 $\pm$.01</u>	0.25 $\pm$.02	0.30 $\pm$.02	0.28 $\pm$.01	0.30 $\pm$.04	-2.3
Muscle binding	0.26 $\pm$.01	0.23 $\pm$.02	<u>0.25 $\pm$.04</u>	0.27 $\pm$.03	0.40 $\pm$.02	-10.1
Mouse liver clearance	0.34 $\pm$.03	0.33 $\pm$.02	<u>0.34 $\pm$.03</u>	0.37 $\pm$.01	0.35 $\pm$.02	-1.9
Plasma binding	0.24 $\pm$.03	<u>0.25 $\pm$.02</u>	0.29 $\pm$.01	0.30 $\pm$.00	0.29 $\pm$.01	+0.9
PXR						
PXR activity	0.49 $\pm$.03	<u>0.51 $\pm$.01</u>	0.55 $\pm$.03	0.52 $\pm$.06	0.52 $\pm$.03	+3.2
Mean	0.37 $\pm$.00	<u>0.38 $\pm$.01</u>	0.41 $\pm$.01	0.41 $\pm$.01	0.44 $\pm$.02	

Overlap between ChEMBL adaptation and OpenADMET

The ChEMBL adaptation stage and OpenADMET contain a modest number of shared molecular structures. We compared the 14,261 unique standardized molecules in OpenADMET-23 against the 511,886 valid standardized molecules used during ChEMBL adaptation. Molecules were standardized using molecular cleanup, fragment-parent selection, uncharging, and sanitization.

We considered four increasingly broad definitions of exposure: (i) identical standardized structures; (ii) identical molecular connectivity, measured using the connectivity block of the InChIKey and therefore ignoring stereochemistry and protonation; and ECFP4 neighbours with Tanimoto similarity (iii) ≥ 0.90 or (iv) ≥ 0.70 . These definitions provide progressively broader notions of structural exposure, but they are evaluated independently and their counts should not be added together.

Structural matching alone does not establish whether a ChEMBL assay is semantically related to a downstream OpenADMET endpoint. During adaptation, matched molecules are associated with labels from the 642 selected ChEMBL binding or functional assays rather than with the

Table A3. Paired baseline contrasts. Differences and 95% bootstrap intervals use 23 endpoints. Wilcoxon and sign-test p -values are Holm-adjusted within separate six-comparison families.

Comparator	FT minus baseline	95% CI	Wilcoxon p	Sign p
ECFP4-LightGBM	-0.0334	[-0.0587, -0.0116]	0.0060	0.0931
RDKit-LightGBM	-0.0377	[-0.0590, -0.0192]	0.00082	0.00026
RDKit random forest	-0.0564	[-0.0813, -0.0348]	0.000063	0.00026
Chemprop	-0.0654	[-0.1008, -0.0375]	0.000030	0.000029
ECFP4 random forest	-0.0783	[-0.1114, -0.0490]	0.000041	0.00098
Training median	-0.2177	[-0.2845, -0.1612]	0.0000014	0.0000014

Table A4. OpenADMET dataset and split accounting. Positive-log transforms are applied before modeling; identity denotes the retained source target. Test sizes correspond to the three stored outer cluster assignments; the remaining outer-training rows are divided into inner train and 15% cluster-held-out validation sets.

Family	Endpoint	Rows	Target transform	Test rows (splits 1/2/3)
ASAP	HLM	299	positive log ₁₀	32/23/23
ASAP	KSOL	365	positive log ₁₀	53/48/49
ASAP	LogD	352	identity	55/69/59
ASAP	MDR1 efflux	424	positive log ₁₀	56/65/51
ASAP	MERS-CoV potency	901	identity	81/118/118
ASAP	MLM	300	positive log ₁₀	62/51/70
ASAP	SARS-CoV-2 potency	842	identity	69/116/50
Biogen ADME	HLM CLint	3,087	identity	104/75/75
Biogen ADME	HPPB	194	identity	25/20/21
Biogen ADME	MDR1 efflux	2,642	identity	77/75/76
Biogen ADME	RLM CLint	3,054	identity	107/124/104
Biogen ADME	RPPB	168	identity	20/20/20
Biogen ADME	solubility	2,173	identity	98/113/125
ExpansionRx	Caco-2 efflux	2,161	positive log ₁₀	339/260/208
ExpansionRx	Caco-2 permeability	2,156	positive log ₁₀	238/214/202
ExpansionRx	HLM CLint	3,595	positive log ₁₀	443/409/426
ExpansionRx	KSOL	5,128	positive log ₁₀	608/496/520
ExpansionRx	LogD	5,039	identity	534/570/551
ExpansionRx	MBPB	973	positive log ₁₀	120/113/176
ExpansionRx	MGMB	221	positive log ₁₀	15/22/44
ExpansionRx	MLM CLint	4,375	positive log ₁₀	617/528/655
ExpansionRx	MPPB	1,292	positive log ₁₀	133/147/146
PXR	PXR activity	4,140	identity	237/225/228

OpenADMET records used for downstream evaluation. Nevertheless, structural exposure, and potentially related biological supervision, could make a matched molecule easier to transfer. We therefore test whether structurally exposed molecules account for the reported OpenADMET performance.

We removed exposed molecules from each test partition and recomputed MAE using the stored predictions. Models were not retrained, and the same row filters were applied to GRASP and ECFP4-LightGBM. The original GRASP mean endpoint MAE was 0.3742. Removing exact matches reduced it to 0.3724, while removing every test molecule with an ECFP4 neighbour of similarity at least 0.70 reduced it to 0.3693. Thus, performance did not deteriorate when previously seen molecules were excluded.

GRASP also retained its advantage over ECFP4-LightGBM under every exclusion rule: the filtered mean MAE difference ranged from -0.0330 to -0.0355, with GRASP winning on 16–17 of the 23 endpoints. We therefore find no evidence that the reported OpenADMET advantage is caused by direct molecule exposure during ChEMBL adaptation. This analysis tests structural exposure; it does not determine whether individual ChEMBL assays measure properties semantically

Table A5. Executed OpenADMET downstream configuration. All three GRASP adaptation modes use a 512-unit endpoint head, AdamW, Smooth-L1 training loss, a 10% linear warmup, gradient clipping at 1.0, and validation-MAE early stopping.

Setting	Frozen mix	LoRA mix	Full mix
Learning rate	10^{-3}	5×10^{-5}	10^{-5}
Epoch cap / patience		60 / 12 epochs	
Batch size / weight decay		64 / 10^{-4}	
Precision	bfloat16 when CUDA is available		
Mixed layers		4, 8, 10, 12	
LoRA rank / α / dropout	–	16 / 32 / 0.1	–
Trainable encoder	no	query/value LoRA	yes

Table A6. Adaptation parameter counts verified across all 69 endpoint-split artifacts. LoRA adds adapter parameters, so its total parameter count differs slightly from the base encoder.

Method	Trainable	Total	Trainable fraction
Full fine-tuning	93,881,350	93,881,350	100.00%
LoRA query/value	986,374	94,471,174	1.04%
Frozen layer mixing	396,550	93,881,350	0.42%

related to particular OpenADMET endpoints.

Table A7. OpenADMET adaptation by endpoint (mean MAE across three reconstructed cluster splits; lower is better). FT updates the full encoder, frozen mixing learns a readout over fixed encoder layers, and LoRA updates low-rank query/value adapters.

Endpoint	FT	Frozen mixing	LoRA
Caco-2 permeability	0.294 $\pm$ 0.018	0.301	0.297
Caco-2 efflux	0.233 $\pm$ 0.010	0.233	0.231
ExpansionRx LogD	0.272 $\pm$ 0.017	0.312	0.306
ExpansionRx KSOL	0.416 $\pm$ 0.024	0.468	0.415
ExpansionRx HLM	0.350 $\pm$ 0.024	0.366	0.369
ExpansionRx MLM	0.340 $\pm$ 0.026	0.354	0.364
MBPB	0.250 $\pm$ 0.010	0.253	0.255
MGMB	0.258 $\pm$ 0.011	0.290	0.290
MPPB	0.244 $\pm$ 0.028	0.265	0.256
MERS-CoV potency	0.557 $\pm$ 0.109	0.609	0.576
SARS-CoV-2 potency	0.617 $\pm$ 0.050	0.611	0.631
ASAP LogD	0.471 $\pm$ 0.064	0.457	0.454
ASAP KSOL	0.315 $\pm$ 0.045	0.331	0.334
ASAP HLM	0.355 $\pm$ 0.072	0.308	0.330
ASAP MLM	0.491 $\pm$ 0.050	0.433	0.477
ASAP MDR1	0.323 $\pm$ 0.030	0.346	0.333
PXR activity	0.493 $\pm$ 0.032	0.508	0.483
Biogen solubility	0.373 $\pm$ 0.003	0.385	0.369
Biogen HLM	0.327 $\pm$ 0.008	0.327	0.333
Biogen RLM	0.376 $\pm$ 0.024	0.401	0.377
Biogen HPPB	0.469 $\pm$ 0.133	0.476	0.450
Biogen RPPB	0.445 $\pm$ 0.060	0.486	0.480
Biogen MDR1	0.335 $\pm$ 0.023	0.380	0.329
Mean, all 23	0.374 $\pm$ 0.003	0.387	0.380

Table A8. Protocol-matched frozen-encoder comparison on OpenADMET-23 (lower is better). Mol-JEPA uses its released frozen CLS representation, while GRASP uses its frozen encoder with learned layer mixing and atom-attention pooling. Both representations use the same regression MLP and the same three reconstructed cluster-held-out splits. **Bold** marks the lower MAE and underlining marks the other result. $\Delta\% = 100(\text{Mol-JEPA} - \text{GRASP})/\text{Mol-JEPA}$; positive values favor GRASP.

Source	Endpoint	Mol-JEPA	GRASP	$\Delta\%$
ASAP	Human liver clearance	<u>0.410</u>	0.308	+24.9
	Solubility	<u>0.393</u>	0.331	+15.9
	LogD	<u>0.545</u>	0.457	+16.1
	MDR1 efflux	<u>0.378</u>	0.346	+8.5
	MERS-CoV potency	<u>0.619</u>	0.609	+1.7
	Mouse liver clearance	0.409	<u>0.433</u>	-5.9
	SARS-CoV-2 potency	<u>0.691</u>	0.611	+11.6
Biogen	Human liver clearance	<u>0.354</u>	0.327	+7.6
	Human plasma binding	<u>0.564</u>	0.476	+15.6
	MDR1 efflux	<u>0.384</u>	0.380	+1.1
	Rat liver clearance	<u>0.411</u>	0.401	+2.3
	Rat plasma binding	0.448	<u>0.486</u>	-8.3
	Solubility	0.367	<u>0.385</u>	-5.1
ExpansionRx	Caco-2 efflux	0.227	<u>0.233</u>	-2.8
	Caco-2 permeability	<u>0.340</u>	0.301	+11.6
	Human liver clearance	<u>0.376</u>	0.366	+2.7
	Solubility	<u>0.492</u>	0.468	+5.0
	LogD	<u>0.371</u>	0.312	+15.8
	Brain binding	<u>0.286</u>	0.253	+11.4
	Muscle binding	0.216	<u>0.290</u>	-34.5
	Mouse liver clearance	<u>0.369</u>	0.354	+3.9
	Plasma binding	<u>0.295</u>	0.265	+10.2
PXR	PXR activity	<u>0.539</u>	0.508	+5.8
Mean		<u>0.4123</u>	0.3869	+6.2
Favorable endpoints		5/23	18/23	—

Table A9. Structural overlap between OpenADMET-23 and the ChEMBL adaptation data. OpenADMET contains 14,261 unique standardized molecules and 43,881 labelled endpoint rows. "Test rows" pools the 12,021 test instances across the 23 endpoints and three cluster-held-out splits. A molecule can occur in more than one endpoint, so molecule and row counts differ.

Overlap rule	Unique molecules		Labelled rows		Test rows	
	Count	Fraction	Count	Fraction	Count	Fraction
Standardized exact	577	4.05%	1,110	2.53%	203	1.69%
Same connectivity	662	4.64%	1,280	2.92%	219	1.82%
ECFP4 Tanimoto ≥ 0.90	674	4.73%	1,301	2.96%	228	1.90%
ECFP4 Tanimoto ≥ 0.70	1,174	8.23%	2,500	5.70%	428	3.56%

Table A10. OpenADMET performance after excluding molecules represented during ChEMBL adaptation. Δ_{filter} is filtered minus original GRASP MAE; negative values indicate lower error after exclusion. The confidence interval was obtained by bootstrapping the 23 endpoint-level differences 100,000 times. The GRASP-ECFP4 column is evaluated on the same retained rows; negative values favor GRASP.

Exclusion rule	Endpoints exposed	Filtered MAE	Δ_{filter}	95% CI	GRASP-ECFP4
Standardized exact	13/23	0.3724	-0.0018	[-0.0062, +0.0019]	-0.0333
Same connectivity	14/23	0.3721	-0.0021	[-0.0065, +0.0016]	-0.0336
ECFP4 Tanimoto ≥ 0.90	17/23	0.3724	-0.0018	[-0.0064, +0.0025]	-0.0330
ECFP4 Tanimoto ≥ 0.70	22/23	0.3693	-0.0049	[-0.0131, +0.0017]	-0.0355

Table A11. TDC ADMET evaluation tasks. N is the total number of labelled rows across the fixed train-validation pool and test set. AUROC, AUPRC, and Spearman correlation are maximized; MAE is minimized.

Task	Type	Metric	N
Ames	Classification	AUROC	7,278
BBB	Classification	AUROC	2,030
Bioavailability	Classification	AUROC	640
CYP3A4 substrate	Classification	AUROC	670
DILI	Classification	AUROC	475
HIA	Classification	AUROC	578
P-glycoprotein	Classification	AUROC	1,218
hERG	Classification	AUROC	655
CYP2C9 inhibition	Classification	AUPRC	12,092
CYP2C9 substrate	Classification	AUPRC	669
CYP2D6 inhibition	Classification	AUPRC	13,130
CYP2D6 substrate	Classification	AUPRC	667
CYP3A4 inhibition	Classification	AUPRC	12,328
Caco-2 permeability	Regression	MAE	910
LD50	Regression	MAE	7,385
Lipophilicity	Regression	MAE	4,200
Plasma protein binding	Regression	MAE	2,790
Solubility	Regression	MAE	9,982
Hepatocyte clearance	Regression	Spearman	1,213
Microsomal clearance	Regression	Spearman	1,102
Half-life	Regression	Spearman	667
Volume of distribution	Regression	Spearman	1,130

TDC Data, Splits, and Evaluation

Data. We evaluate molecular-property transfer using the Therapeutics Data Commons (TDC) ADMET benchmark group [15]. The benchmark contains 22 endpoint-specific datasets spanning absorption, distribution, metabolism, excretion, and toxicity. Thirteen endpoints are binary classification tasks and nine are regression tasks. We retain the task-specific evaluation metric specified by TDC: AUROC for eight classification tasks, AUPRC for five classification tasks, MAE for five regression tasks, and Spearman correlation for four regression tasks. Table A11 lists the complete evaluation panel.

The 22 datasets contain 81,809 labelled endpoint-molecule rows in total. This is a sum over endpoints rather than a count of unique molecules because the same molecule can occur in more than one endpoint dataset. Labels are used on the scale supplied by TDC; we do not standardize regression targets or apply additional target transformations.

All SMILES in the evaluated benchmark files are RDKit-valid. The downstream loader does not apply a global heavy-atom filter and does not truncate molecules. Molecules larger than the 96-heavy-atom limit used to construct the pretraining data are processed as supplied, provided that they fit within the encoder’s positional capacity. Across the 22 task files, 70 endpoint-level rows contain more than 96 heavy atoms. Thus, the 96-heavy-atom threshold is a pretraining-data restriction rather than a downstream TDC exclusion rule.

Splits. For every endpoint, we use the fixed test set distributed with the TDC ADMET_Group benchmark. This test membership is held constant across the final-model GRASP evaluation runs. The historical frozen-encoder comparison is described separately in section*. The test set is not used for hyperparameter selection, checkpoint selection, or early stopping.

TDC provides a combined training-validation pool in addition to its fixed test set. We ap-

ply TDC’s default scaffold-splitting procedure to this pool with seeds 0–4, producing five independently seeded scaffold train–validation partitions. The nominal division of the training–validation pool is 87.5% training and 12.5% validation, corresponding to a nominal overall 70/10/20 train/validation/test allocation. Exact proportions can differ because complete molecular scaffolds are assigned together. These are five scaffold-resampled validation partitions, not five mutually exclusive folds of conventional cross-validation.

For each endpoint, learning rate and dropout are selected using only the mean validation performance across these five partitions. The search space is

$$\text{learning rate} \in \{10^{-5}, 3 \times 10^{-6}, 10^{-6}\}, \quad \text{dropout} \in \{0, 0.10, 0.15\}.$$

The selected configuration maximizes the endpoint’s validation metric. For MAE tasks, this is implemented by maximizing negative MAE. Consequently, each endpoint requires $3 \times 3 \times 5 = 45$ validation runs before its final test configuration is fixed.

Model and optimization. Molecules are represented using the same radius-0 atom-environment tokenization used by GRASP. A [CLS] token is added to each molecular sequence. The final encoder-layer representation of this token is passed through a task-specific dropout–linear–GELU–dropout–linear prediction head. Unless a result is explicitly identified as a frozen-encoder experiment, the complete encoder and prediction head are fine-tuned jointly.

Classification models are optimized with binary cross-entropy on the logits. Regression models are optimized with mean squared error, irrespective of whether their reporting metric is MAE or Spearman correlation. We use Adam with zero weight decay, a batch size of 32, and gradient-norm clipping at 1.0. Training runs for at most 100 epochs. The learning rate increases linearly during the first 10% of optimization steps and remains constant after warmup. Validation is performed every five epochs, and the checkpoint with the best task-specific validation metric is retained.

Locked-test evaluation. After the learning rate and dropout have been fixed, we run three independent downstream training runs with seeds 100, 101, and 102. These runs use scaffold train–validation partitions generated with split seeds 0, 1, and 2, respectively, while preserving exactly the same fixed test membership. The test set is evaluated only after the best validation checkpoint for a run has been selected.

For each endpoint, we report the arithmetic mean and population standard deviation across the three test runs,

$$\bar{s} = \frac{1}{3} \sum_{i=1}^3 s_i, \quad \sigma = \sqrt{\frac{1}{3} \sum_{i=1}^3 (s_i - \bar{s})^2}.$$

The reported deviation therefore reflects both downstream-training randomness and variation among the three scaffold train–validation assignments; it is not pretraining-seed uncertainty.

AUROC, AUPRC, and Spearman correlation are reported with higher values indicating better performance, whereas lower MAE is better. Because the endpoints use different metrics and, for MAE, different target scales, we do not compute a numerical average across the 22 TDC tasks. Comparisons are instead summarized using individual task results and the number of tasks on which each method has the favorable point estimate. ClinTox is not part of this 22-task evaluation panel and is excluded from its task counts.

Table A12. RTD versus MLM on TDC in the controlled 415M-molecule configuration. Values are mean $\pm$ population standard deviation over three seeds. **Bold** marks the favorable point estimate. BBB is excluded because a valid MLM result was not available.

Task	Metric	RTD-15%-S2	MLM-15%-S2
Absorption			
Caco-2	MAE $\downarrow$	0.35 $\pm$.02	0.36 $\pm$.03
Bioavailability	AUROC $\uparrow$	0.68 $\pm$.02	0.63 $\pm$.04
Lipophilicity	MAE $\downarrow$	0.46 $\pm$.01	0.51 $\pm$.01
Solubility	MAE $\downarrow$	0.80 $\pm$.02	0.81 $\pm$.01
HIA	AUROC $\uparrow$	0.97 $\pm$.01	0.96 $\pm$.02
P-glycoprotein	AUROC $\uparrow$	0.92 $\pm$.01	0.93 $\pm$.01
Distribution			
BBB Martins	AUROC $\uparrow$	0.88 $\pm$.02	0.86 $\pm$.02
PPBR	MAE $\downarrow$	7.17 $\pm$.28	7.49 $\pm$.15
VDss	Spearman $\uparrow$	0.56 $\pm$.10	0.50 $\pm$.14
Metabolism			
CYP2C9 inhibition	AUPRC $\uparrow$	0.77 $\pm$.02	0.74 $\pm$.03
CYP2D6 inhibition	AUPRC $\uparrow$	0.69 $\pm$.01	0.63 $\pm$.02
CYP3A4 inhibition	AUPRC $\uparrow$	0.87 $\pm$.01	0.86 $\pm$.01
CYP2C9 substrate	AUPRC $\uparrow$	0.32 $\pm$.02	0.40 $\pm$.03
CYP2D6 substrate	AUPRC $\uparrow$	0.54 $\pm$.04	0.57 $\pm$.04
CYP3A4 substrate	AUROC $\uparrow$	0.65 $\pm$.01	0.67 $\pm$.03
Excretion			
Half-life	Spearman $\uparrow$	0.29 $\pm$.09	0.19 $\pm$.22
Hepatocyte clearance	Spearman $\uparrow$	0.43 $\pm$.02	0.38 $\pm$.05
Microsomal clearance	Spearman $\uparrow$	0.59 $\pm$.02	0.47 $\pm$.06
Toxicity			
LD50	MAE $\downarrow$	0.54 $\pm$.01	0.55 $\pm$.02
hERG	AUROC $\uparrow$	0.82 $\pm$.02	0.75 $\pm$.01
Ames	AUROC $\uparrow$	0.80 $\pm$.01	0.79 $\pm$.01
DILI	AUROC $\uparrow$	0.87 $\pm$.01	0.88 $\pm$.01
Favorable tasks	Count	17/22	5/22